



Appendix: Technical Description NIBP2020 UP with SpO₂ (Variant 3)



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Revision History of this Document

Doc.-Rev.	Date	Author	Remarks
A	01.06.2026	A. Gürbüz	Creation

1 Purpose of the document

This document is intended for professional technicians / engineers who integrate the module into a host system.

This document describes and defines specifications for the medical device NIBP OEM Blood Pressure Board (short: NIBP Module) with SpO₂ functionality.

The following variants are covered by this Technical Description:

NIBP2020 UP HW B+ FW 6.4c

This document contains additional information about SpO₂ functionality. For the NIBP functionality, please refer to the main document.

1.1 Intended Purpose

The NIBP2020 UP module can be provided with a module and the respective sensor for the determination of the oxygen saturation in human beings (minimum weight 4 kg) by means of pulse oximetry (short: SpO₂ module). The module can be reused on multiple patients.

1.2 Features of the NIBP2020 UP with SpO₂ functionality

- The NIBP2020 UP module with SpO₂ functionality features pulse oximetry technology in a very small and low powered design.
- The board consists of a multilayer PCB with surface-mounted components with a total size of 80 mm × 60 mm × 33 mm (module with round pump).
- The SpO₂ module provides oxygen saturation, pulse rate, respiration rate, perfusion index, perfusion variability index, status information, pulse waveform and other output information via the serial digital interface
- Several types of finger sensors available.

1.3 Serious Incidents

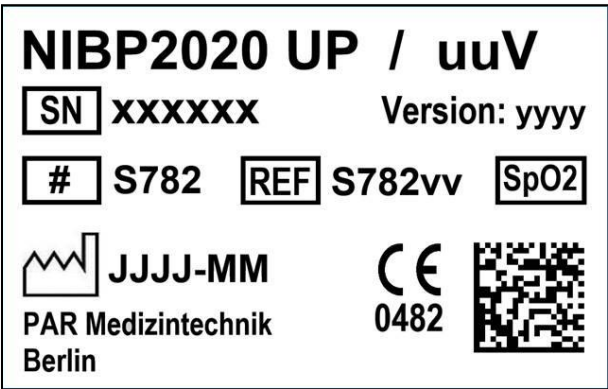
Any serious incident occurring in relation to the device should be reported to the manufacturer and the competent authority of the Member State in which the user and/or patient is established.

2 Technical Data (Specifications)

The table below contains specifications that are different from the modules without SpO₂ module.

2.1 NIBP part of NIBP2020 UP with SpO₂ functionality

Mechanical data: (Module with flat pump)	Module dimensions: 80 mm × 60 mm × max 25 mm (l × w × h) Weight: 100 g Deflation mode only See also
Mechanical data: (Module with round pump)	Module dimensions: 80 mm × 60 mm × max 33 mm (l × w × h) Weight: 120 g See also
Baud rate	8N1 with 19200 baud, hardware reset

Design	Explanation of the Symbols	
		NIBP2020 UP with additional SpO₂ module. For the other symbols, see the main document.
uu, xxxxxx, yyyy, vv, and JJJJ-MM are place holders for the voltage, serial number, version number, article number and manufacturing date, respectively. This label is attached to the pump of the board and on the protection bag.		

2.2 Pulse Oximetry Part of NIBP2020 UP with SpO₂ functionality

O ₂ saturation range:	35 – 100 %
O ₂ saturation accuracy:	70 ... 79 %: $A_{rms} \leq 3 \%$ 80 ... 100 %: $A_{rms} \leq 2 \%$ < 70 %: accuracy unspecified Invalid value: 0x7F
Heart rate / pulse rate (PR):	Measurement range: 25 – 250 bpm Accuracy: ± 2 bpm Resolution: 1 bpm Invalid value: 0xFF
Perfusion Index (PI):	Measurement range: 1 – 200 % Measurement Accuracy: not defined Invalid Value: 0
Status information:	for details see section 7.1 Status information is detected in the module and reported to the monitor via the communication link.
Transmission:	See Chapter 7.1.
Connector:	MiniMed
Temperature:	0 °C to 40 °C (operating) -20 °C to 60 °C (transport and storage)
Humidity:	30 % - 95 % (operating) 10 % - 100 % (transport and storage)
Atmospheric Pressure:	860– 1060 kPa (operating) 700 – 1060 kPa (transport and storage)
Wavelengths (Finger sensor):	Red: approximately 660 nm Infrared: approximately 905 nm

2.3 Disposal

The module must not be disposed of with the household waste, but separately with the electronic waste disposal. For information on the disposal of the storage medium, please contact the customer service of PAR Medizintechnik GmbH & Co. KG (service@par-berlin.com).

WEEE-Reg.-Nr.: DE63208995

3 Transport and Storage Conditions

Temperature range: -25 °C to +55 °C

Humidity: 10 – 100%

4 Standards

Standard for SpO₂ functionality:

ISO 80601-2-61 (SpO₂ measurement)

5 Protocol Direction from Monitor to the NIBP module

This section describes differences in the commands to be sent from monitor to the host. They supersede the commands from the main document.

5.1 Frame Scheme

Table 1: Format of a Frame

Char 1	Char 2	Char 3	Char 4	Char 5	Char 6	Char 7	Char 8
<STX>	c ₀	c ₁	;	;	x ₀	x ₁	<ETX>

Table 2: Explanation of the Frame scheme

<STX> (Start of Text)	= 0xFD
c ₀ and c ₁	= command code (2 ASCII characters. Range of values from 00 to 99)
; and ;	= 2 times semicolon
x ₀ and x ₁	= checksum (2 ASCII characters)
<ETX> (End of Text)	= 0xFE

Table 3: Example for command code 01 (Start BP measurement)

Command Code	<STX>	0	1	;	;	D	7	<ETX>
Hex Notation	0xFD	0x30	0x31	0x3B	0x3B	0x44	0x37	0xFE

5.2 Command codes for NIBP2020 UP with SpO₂ functionality

The following commands apply for NIBP2020 UP with SpO₂ functionality. They supersede the commands from the main document.

Command Code	Checksum	Function
60	DC	Set start pressure to 080 mmHg (only adult)
61	DD	Set start pressure to 100 mmHg (only adult)
62	DE	Set start pressure to 120 mmHg (only adult)
30	D9	SpO ₂ data stream off
31	DA	SpO ₂ data stream on
92	E1	SpO ₂ plethysmogram off
93	E2	SpO ₂ plethysmogram on
94	E3	Get Firmware version
95	E4	Get Hardware version

6 Direction from the NIBP module to Monitor

6.1 Basic frame structure:

<STX> Message bytes <ETX><CR>

<STX> (Start of Text) = 0xFD

<ETX> (End of Text) = 0xFE

<CR> (Carriage Return) = 0x0D

7 SpO₂ functionality

The SpO₂ module together with the SpO₂ sensor provides the SpO₂ functionality for the NIBP module.

7.1 Data Structure

The following tables display the format of a data packets sent from the module ().

- **Serial port settings**

Baud rate	Data bits	Parity	Stop bits	Flow control
9600	8	None	1	None

- **Serial data packet format**

Packet header	Packet length	Packet body	Checksum
0x55 0xAA (2Bytes)	N (1Byte)	A1, A2, A3, ...An (n Bytes)	Sum (1Byte)

Description:

Packet Header: 2 Bytes fixed value (0x55 0xaa)

Packet Length: The total number of remaining bytes in the data packet excluding the packet header.

$$N = n+2 \text{ ----- } \{n \text{ Bytes (Packet body)} + 1 \text{ Byte (Packet length)} + 1 \text{ Byte (Checksum)}\}$$

Packet body: The measurement data in the data packet. See below for details.

Checksum: The packet checksum. $\text{Sum} = (N+A1+A2+\dots+An) \% 256$, % is mod operator.

● **Packet body format**

■ **Packet body 1#(Transfer Frequency @ 1Hz)**

A1	A2	A3	A4	A5
Packet Index	Status	SpO2	PR	PI

Description:

Packet Index: The index of the current packet, range (0~255), This is mainly used for packet loss detection.

Status: The measurement status of the module, composed of the following flags.

0x01 Sensor Off --- SpO2 sensor is disconnected

0x02 No Finger --- Finger is not detected

0x04 No Pulse --- Pulse signal is not found

0x10 Searching --- Searching for pulse signal

0x40 Signal Weak --- Pulse signal is too weak

SpO2: Blood oxygen saturation measurement value, range (0~100), invalid value=127(0x7f), unit : %

PR: Pulse rate measurement value, range (25~250), invalid value=255(0xff), unit : %

PI: Blood perfusion index measurement value, range (1~200), invalid value=0, unit : %

■ **Packet body 2# (Transfer Frequency @ 40Hz)**

A1	A2	A3	A4	A5	A6	A7	A8
Packet Index	Pulse flag	Low byte of ADCSa mple1	Middle byte of ADCSa mple1	High byte of ADCSa mple1	Low byte of ADCSa mple2	Middle byte of ADCSa mple2	High byte of ADCSa mple2

A9	A10	A11	A12	A13	A14	A15	A16	A17
Low byte of ADCSa mple3	Middle byte of ADCSa mple3	High byte of ADCSa mple3	Low byte of ADCSa mple4	Middle byte of ADCSa mple4	High byte of ADCSa mple4	Low byte of ADCSa mple5	Middle byte of ADCSa mple5	High byte of ADCSa mple5

Description:

Packet Index: The index of the current packet, range (0~255). This is mainly used for packet loss detection.

Pulse flag: Pulse Peak Search Flag Bit.

0x00	Pulse peak is searching.
0x01	Pulse peak is found at position of ADCSample1 (i.e. Heart Beat Tip)
0x02	Pulse peak is found at position of ADCSample2 (i.e. Heart Beat Tip)
0x04	Pulse peak is found at position of ADCSample3 (i.e. Heart Beat Tip)
0x08	Pulse peak is found at position of ADCSample4 (i.e. Heart Beat Tip)
0x10	Pulse peak is found at position of ADCSample5 (i.e. Heart Beat Tip)

ADCSample1~ADCSample5: The values of 5 consecutive PPG waveform sampling points.

Every value range is 0~818800(0xC7E70), saved by 3 bytes.

ADCSample1 is the oldest value, and ADCSample5 is the latest value.

7.2 Example

Below you can see a possible extract from the data stream of the NIBP module to the host monitor.

Example of firmware version response package (17 bytes):

0x55 0xaa 0x0f 0x46 0x56 0x31 0x2e 0x30 0x34 0x2e 0x30 0x30 0x2e 0x46 0x46 0x00 0xb6
 0x55 0xaa ----- Packet head
 0x0f ----- Packet length
 0x46 ----- ASCII 'F', mark of firmware version
 0x56 0x31 0x2e 0x30 0x34 0x2e 0x30 0x30 0x2e 0x46 0x46 ----- firmware version string
 'V1.04.00.FF'
 0xb6 ----- Checksum

Example of hardware version response package (17 bytes):

0x55 0xaa 0x0f 0x48 0x56 0x31 0x2e 0x34 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x40
 0x55 0xaa ----- Packet head
 0x0f ----- Packet length
 0x48 ----- ASCII 'H', mark of hardware version
 0x56 0x31 0x2e 0x34 ----- hardware version string 'V1.4'
 0x40 ----- Checksum

Note 1: "Packet Body #2" is sent with a frequency of 40 Hz, "Packet Body #1" with 1 Hz.

Note 2: "Packet Body 2#" (plethysmogram) is not sent when auscultatory measurement feature is used.

7.3 Application of the SpO₂ Sensor

7.3.1 Intended Use SpO₂ Sensor

The SpO₂ sensors are used for patient monitors and pulse oximeters in producers and medical units. They are applicable to adult (weight:>30kg) and pediatric (weight:10~50kg) respectively for checking continuous noninvasive measurement of arterial oxygen saturation and pulse rate.

NOTICE: The SpO₂ sensor is applicable to adult and pediatric patients respectively for spot check or continuous noninvasive measurement of arterial oxygen saturation and pulse rate. BST09001S SpO₂ sensor is only used with BM2000 Pulse oximeter. This sensor must be used by trained clinical professionals. If the product is intended to allow direct diagnosis or monitoring of vital physiological processes, then it is likely to result in immediate danger to the patient).

7.3.2 Contraindications:

When performing continuous prolonged monitoring on a patient, regularly inspect and change the sensor application site according to the skin quality. Inspecting at least every two hours is recommended. The damaged skin tissue can't be measured.

7.3.3 Application

- 1) The best site to be monitored is index finger. Considering other fingers when the index finger is not available or can't be located correctly.
- 2) Place an finger over the monitoring window of the sensor, aligning the fingernail with the nail mark on the surface of the sensor, don't expose the tip of the index finger under measurement.
- 3) Position the cable along the top of the hand.
- 4) Plug the reusable sensor into the oximeter and verify proper operation as described in the oximeter operator's manual.

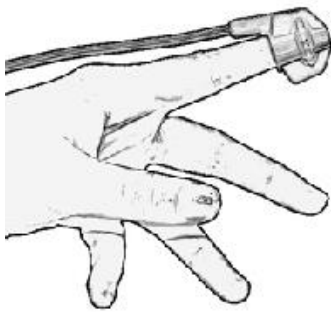


Figure 1: Application of the Silicone SpO₂ Sensor

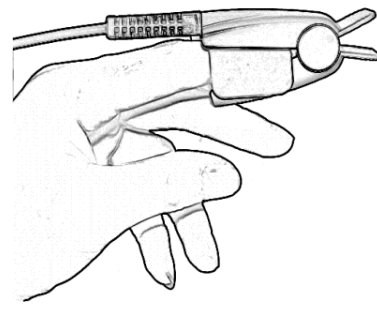


Figure 2: Application of the finger Clip SpO₂ Sensor

NOTE:

- Check the SpO₂ sensor and the cable before using it. Don't use them for SpO₂ monitoring in case of damage, block, deterioration or visible contamination.
- Using the SpO₂ sensor in the presence of bright lights may result in inaccurate measurements. In such cases, cover the sensor site with opaque material.
- Try to keep the patient still (especially the arm) and avoid the measured site suffering excessive motion.
- Don't put the SpO₂ sensor on extremities with arterial catheter or venous syringe.
- Don't perform SpO₂ monitoring and NIBP measurements on the same arm simultaneously. Obstruction of blood flow during NIBP measurements may adversely affect the reading of the SpO₂ value.
- Testee's fingernail can't be too long. Otherwise the finger can't be inserted into the sensor to a suitable depth and the SpO₂ measurements may be inaccurate.
- Always choose a site that is well perfusion and will completely cover the sensor's detector window. Site should be cleaned of debris and dry prior to sensor placement.
- Make sure to place the SpO₂ sensor on the finger in a correct direction. The LED part of the sensor should be at the backside of the patient hand and photodetector part at the inside. Make sure to insert the finger to suitable depth into the sensor so that the fingernail is just opposite to the light emitted from the sensor.
- The SpO₂ sensor must be moved to a new site at least every 2 hours. Because individual skin condition affects the ability of the skin to tolerate sensor placement, it may be necessary to change the sensor site more frequently with some patients. If skin integrity changes, move the sensor to another site.

- A proper sensor shall be selected according to different patients and application sites. Tight sensors may cause venous pulsation, obstructed blood circulation, pressure marks, pressure necrosis, artifacts and inaccurate measurement, while loose sensor may lead to erroneous optical alignment or falling off. If the sensor is too tight because the application site is too large or becomes too large due to edema, excessive pressure may result in venous congestion distal from the application site, leading to interstitial edema and tissue ischemia.
- The highest temperature of the SpO₂ sensor contacts with patient's skin don't be allowed more than 41°C.
- Shock, anemia, hypothermia and the application of vasoconstriction drug may decrease arterial blood flow to an unmeasurable level.
- Pigment, or deep color (for example: nail polish, artificial nails, dye or pigmented cream) may cause inaccurate measurements.

7.4 Cleaning and Disinfection of the SpO₂ Sensor

The instructions for cleaning and disinfection of the SpO₂ sensors follow the instructions of the manufacturer of the SpO₂ sensor.

Note:

- Never immerse or soak the SpO₂ sensor.
- Exercise caution during cleaning/disinfection to avoid wetting the pins.
- We recommend that the SpO₂ sensor be disinfected only when necessary as determined by your hospital's policy, to avoid long term damage to the SpO₂ sensor.
- Never use cleaning agents/disinfectants other than the recommended.
- Do not sterilize by irradiation, steam, or ethylene oxide. Never permit high-pressure and high-temperature disinfection of the SpO₂ sensor.

7.4.1 Cleaning:

- 1) Clean the SpO₂ sensor with cotton or soft cloth moistened with water.
- 2) After cleaning, wipe off the water with a soft cloth.
- 3) Allow the SpO₂ sensor to air dry.

7.4.2 Disinfection

The recommended disinfectants include: ethanol 70%, isopropanol 70%, glutaraldehyde(2%) solution disinfectants.

- 1) Clean the SpO₂ sensor as instructed above.
- 2) Disinfect the SpO₂ sensor with cotton or soft cloth moistened with one of the recommended disinfectants.
- 3) After disinfection, be sure to wipe off the disinfectant left on the SpO₂ sensor with a soft cloth moistened with water.
- 4) Allow the SpO₂ sensor to air dry.

7.5 Warnings and Precautions

The following **WARNINGS** are specified in the instructions for use of the **SpO₂ – sensors**:

- The person who uses the SpO₂ sensor must receive adequate training before use.
- Do not use the SpO₂ sensor during MRI scanning. Conducted current may cause burns. Also, the sensor may affect the MRI image, and the MRI unit may affect the accuracy of oximetry measurements.
- **EXPLOSION HAZARD:** Do not use the SpO₂ sensor in the presence of flammable anesthetics, explosive substances, vapors or liquids.
- Interference of high frequency electricity may affect the accuracy of the measurement.
- The user shall be responsible for ensuring the compatibility between this sensor, the extension cable and other equipment by either referring to the instructions for using this sensor or contacting us before use.
- The user must choose correct size for SpO₂ sensor according to the tested part size before use. Please consult with the manufacturer for specific model.
- No modification of this product is allowed. Maintenance should be operated by professional maintenance personnel who are approved by manufacturers.
- Failure to apply the SpO₂ sensor properly may cause incorrect measurements.
- It's the same as other medical equipment, should connect the cable carefully. And avoid patient is winded or suffocates. Electric surgical equipment cable can't entwine with sensor cable.
- Do not alter or modify the SpO₂ sensor. Alterations or modifications may affect performance or accuracy.
- For disposal of the SpO₂ sensor and accessories, follow local regulations or your hospital's policy regarding disposal of such accessories. Don't dispose randomly.
- Federal law (USA) restricts this device to sale by or on the order of a physician.

7.6 List of SpO₂ sensors and accessories

Article number PAR	Product name
A4051	Adult silicone SpO ₂ sensor
A4052	Adult finger clip SpO ₂ sensor
A2201	Internal connection cable (Molex to MiniMed) Note: Length can be adapted to customer requirements

7.7 Service life

Valid within 5 years of the production date, but it is recommended to replace the device every 6 months after opening. If the device is still needed, please evaluate the performance of the device, and strict inspection should be made during use.

8 Information to be provided in the User manual

- Definition of the user groups, see section 1.1
- How to apply the SpO₂ sensor, see section 7.3
- The operation and storage conditions (see sections 2.2 and 3)
- Intended medical indication and intended use, see section 1.1
- A statement like “A functional tester cannot be used to assess the accuracy of the SpO₂ measurement function reliably. If you are unsure about the accuracy of a measurement, verify the patient’s vital signs using other methods and then ensure that the SpO₂ measurement is functioning correctly.”
- Factors that can affect the performance of the SpO₂ functionality, warnings and precautions: see section 7.5
- the technical and accuracy information from section 2.2
- Cleaning and disinfection, maintenance: See sections 7.4 and 7.7
- List of SpO₂ sensors and accessories, see section 7.6
- Disposal: See section 2.3